

usecase1.ipynb - Colab

vtu30293-cmyk/ihlusecasetasks

Ask Gemini

colab.research.google.com/drive/1U1fP2MvDVdExNEifDPtB5DgvcQ0JS4xh

School

Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle.

Explore the Plans

usecase1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

...

Chatbot: Hello! I am a simple chatbot.
Chatbot: Type 'bye' to exit.
You: hi
Chatbot: Hello! How are you?
You: what is your name
Chatbot: My name is Python Chatbot.
You: what can you do
Chatbot: I can answer simple questions and chat with you.
You: bye
Chatbot: Goodbye! Have a nice day.

Colab paid products - Cancel contracts here

Variables Terminal

usecase3.ipynb - Colab

vtu30293-cmyk/ihlusecasetasks

Ask Gemini

colab.research.google.com/drive/1p-RIZahXyS782SuAFda0oS0GV0aM1ymV

School

Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle.

Explore the Plans

usecase3.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Connect

Libraries loaded successfully!

Training dataset created!

Number of training examples: 7

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

config.json: 100% 762/762 [00:00<00:00, 73.7kB/s]

tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 3.22kB/s]

vocab.json: 100% 1.04M/1.04M [00:00<00:00, 34.0MB/s]

merges.txt: 100% 456k/456k [00:00<00:00, 22.0MB/s]

tokenizer.json: 100% 1.36M/1.36M [00:00<00:00, 28.2MB/s]

model.safetensors: reconstructing file: 100% 353MB / 353MB, 23.2MB/s

model.safetensors: downloading bytes: 312MB, 28.9MB/s

Loading weights: 100% 76/76 [00:00<00:00, 2400.08it/s]

generation_config.json: 100% 124/124 [00:00<00:00, 12.5kB/s]

LLM loaded: distilgpt2

Map: 100% 7/7 [00:00<00:00, 193.43 examples/s]

Dataset tokenized successfully!

Starting fine-tuning...

/usr/local/lib/python3.13/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no accelerator is found, then device pinned

super().__init__(loader)

[transformers] `loss\_type=None` was set in the config but it is unrecognized. Using the default loss: `ForCausalLMLoss`.

[21/21 00:28, Epoch 3/3]

Step Training Loss

Variables

Terminal

usecase3.ipynb - Colab

vtu30293-cmyk/ihlusecasetasks

Ask Gemini

colab.research.google.com/drive/1p-RlZahXyS782SuAFda0oS0GV0aM1ymV

Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle.

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

LLM + RAG + FINE-TUNING

Chatbot is ready!
Ask a question.
Type 'bye' to exit.

[transformers] Passing `generation\_config` together with generation-related arguments=({'temperature', 'max_new_tokens', 'do_sample', 'pad_token_id'}) is deprecated and will [transformers] Both `max\_new\_tokens` (=60) and `max\_length` (=50) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more inf [transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer GPT2Tokenizer. The clean_up_tokenization post-processing step is designed for WordPiece tokenizer

Bot: What is Python?

Retrieved Context:
Python is a programming language used for artificial intelligence, machine learning, data science, web development and automation.

[transformers] Both `max\_new\_tokens` (=60) and `max\_length` (=50) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more inf

Bot: Java is an object-oriented programming language used for enterprise development, enterprise applications and Android development.

Retrieved Context:
Java is an object-oriented programming language used for backend development, enterprise applications and Android development.

[transformers] Both `max\_new\_tokens` (=60) and `max\_length` (=50) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more inf

Bot:

Retrieved Context:
RAG stands for Retrieval Augmented Generation. RAG retrieves relevant information from documents and provides that information to an LLM before generating an answer.

Variables Terminal

usecase3.ipynb - Colab

vtu30293-cmyk/ihlusecasetasks

Ask Gemini

colab.research.google.com/drive/1p-RIZahXyS782SuAFda0oS0GV0aM1ymV

Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle. Explore the Plans

usecase3.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all

Connect

20 0.456398

21 0.428466

Fine-tuning completed!

modules.json: 100% 349/349 [00:00<00:00, 34.9kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 15.3kB/s]

README.md: 100% 10.5k/10.5k [00:00<00:00, 656kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.04kB/s]

config.json: 100% 612/612 [00:00<00:00, 52.9kB/s]

model.safetensors: reconstructing file: 100% 90.9MB / 90.9MB, 8.93MB/s

model.safetensors: downloading bytes: 85.0MB, 8.26MB/s

Loading weights: 100% 103/103 [00:00<00:00, 2261.22it/s]

tokenizer_config.json: 100% 350/350 [00:00<00:00, 43.3kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 18.1MB/s]

tokenizer.json: 100% 466k/466k [00:00<00:00, 24.6MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 13.5kB/s]

config.json: 100% 190/190 [00:00<00:00, 23.9kB/s]

RAG embeddings created!

Fine-tuned LLM ready!

=====

LLM = RAG + FINE-TUNING

Variables

Terminal

20
0.456398
Variables
Terminal